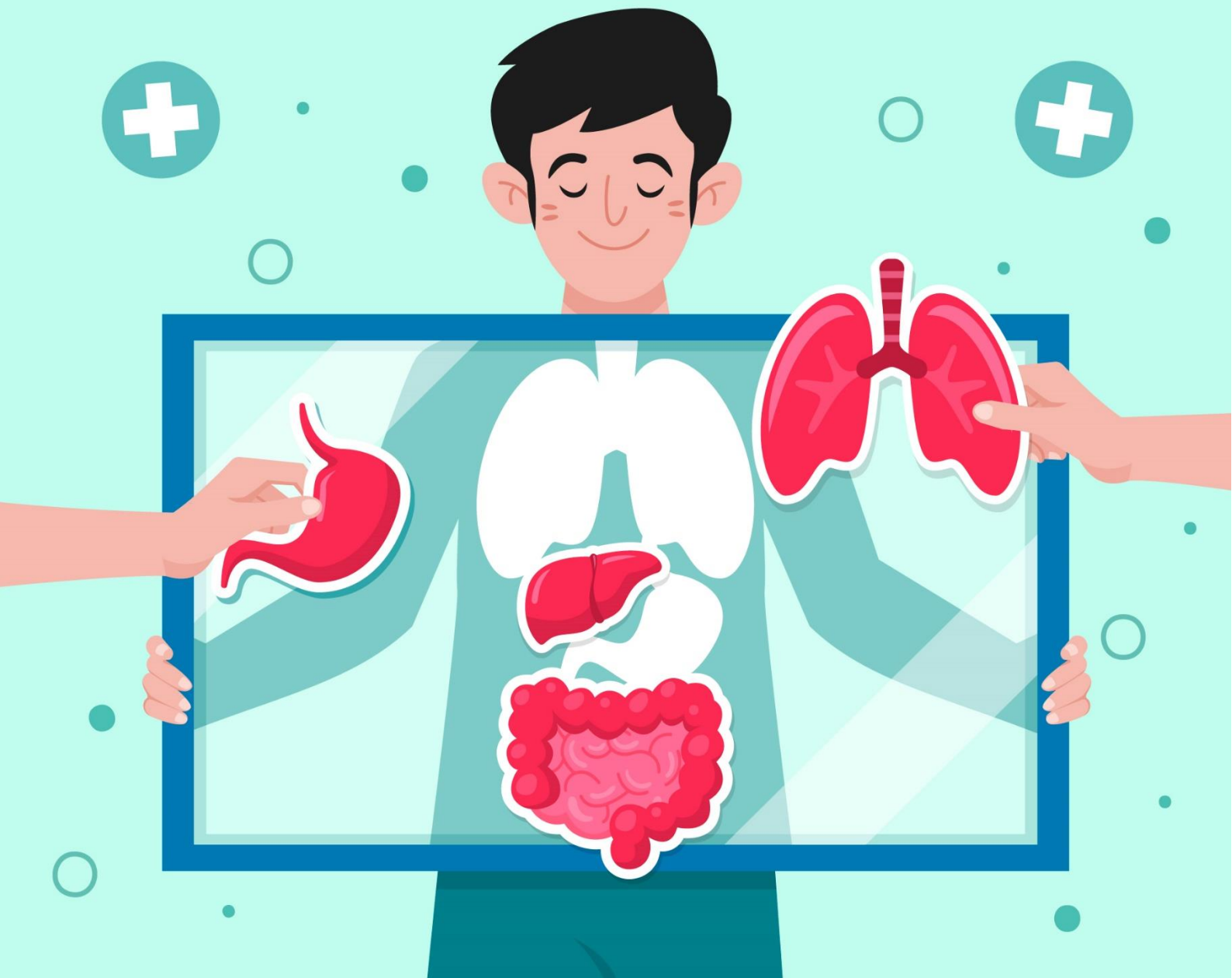




NITI Aayog

DRAFT



SECTOR AT A GLANCE | 11

Organ Donation

Health Sector in India

DRAFT

Sector at a Glance 11

Organ Donation

Health Sector in India

The '**Sector-at-a-Glance**' provides an easy-to-read overview and basic understanding of a sector, sub-sector, or thematic area. The facts and insights on the sector are supported by relevant data and best practices from credible government and multilateral sources. The document covers the structure, key trends, opportunities and challenges, government initiatives, and the way forward for the sector.

Suggested Citation:

NITI Aayog (2023). Organ Donation. NITI Aayog Sector-at-a-Glance Series – Health in India, 11, <link>

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Overview

Organ transplantation is the life-saving therapy for end-stage organ failure. Transplantation involves the removal of damaged/injured tissues or organs from the body of a person and their substitution by similar tissues/organs from a donor; and it is the act of surgical removal of an organ from one person and placing it into another person.

Organ Donation is of significant importance as it directly saves lives, addresses severe organ shortage, and improves recipients' quality of life. The most frequently transplanted solid organs include kidneys, livers, and hearts, and the most frequently transplanted tissues include corneas and musculoskeletal grafts. Organ Donation rates have steadily risen worldwide in the past two decades, delivering remarkable outcomes.¹



Scenario of Organ Scarcity in India²

- Each year, approximately 1.8 lakh individuals experience renal failure in India, yet only around 11,400 renal transplants are conducted annually.
- Liver failure or liver cancer results in loss of lives of an estimated 2 Lakh patients annually, with 10-15% of them potentially being saved through timely liver transplants. Consequently, India requires approximately 25-30 thousand liver transplants annually, but only about 3,700 are being performed.
- Around 50,000 people suffer from heart failure each year, but only around 240 heart transplants take place in India annually. Regarding corneal transplants, approximately 25,000 are performed annually, but the demand is as high as one Lakh.



History and Evolution of Organ Donation in India²

Organ Donation in India has seen significant developments over the years. The first tissue transplant took place in 1948 and the first organ transplant was successfully conducted in 1965. The Government of India introduced The Transplantation of Human Organs Act in 1994 with the aim of regulating removal, storage and transplant of human organs for therapeutic purposes and to address illegal organ trafficking. Brain death was defined in the amendment and was recognised as a valid form of death. Cadaveric organ transplantation began to emerge in the 1980s, leading to India's first successful heart transplant in 1994. Subsequent decades witnessed growth in multi-organ transplants and government initiatives to promote organ donation.

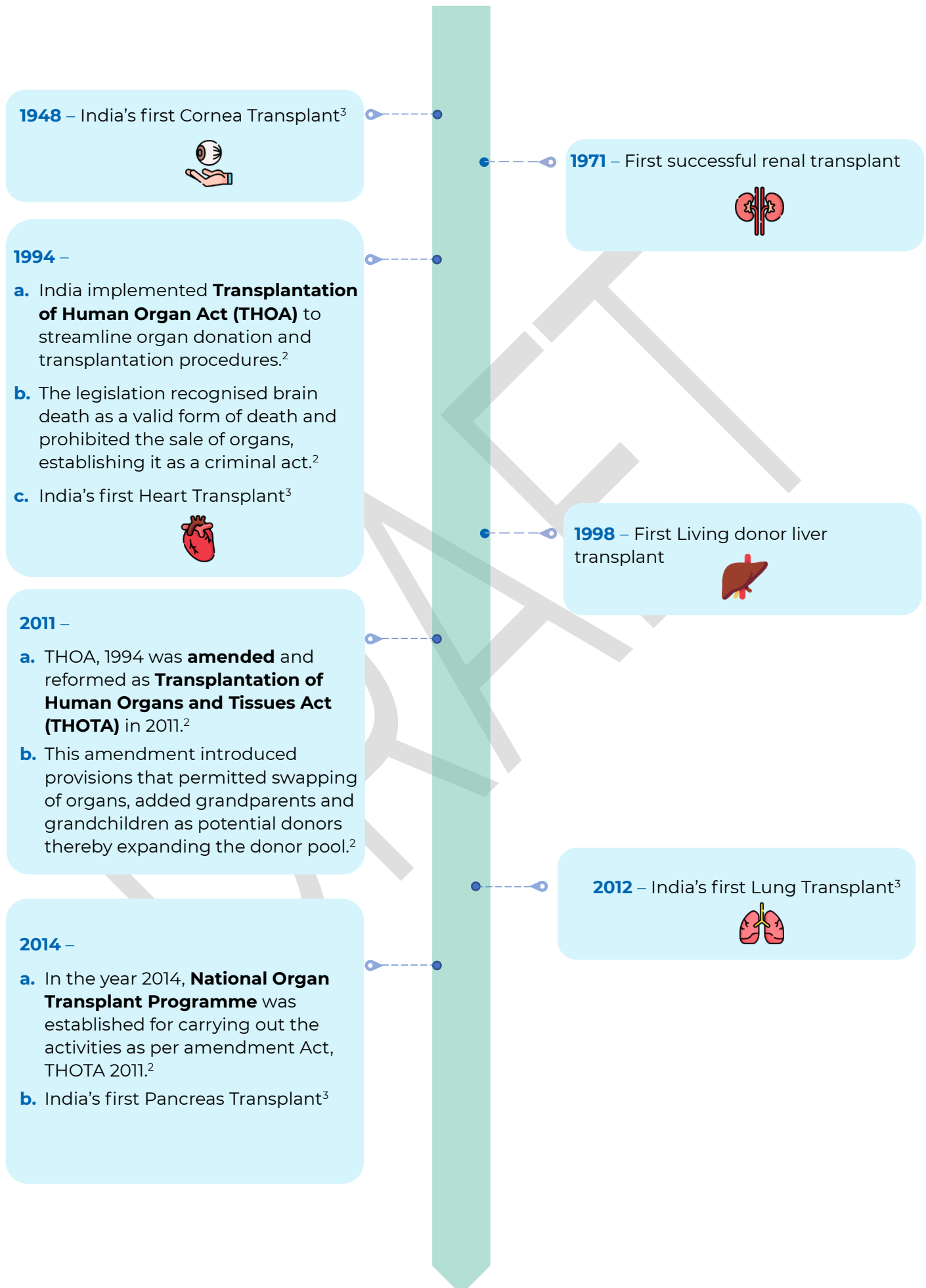
Below is a timeline of key activities under the development of Organ and Tissue Transplantation in India:

NOTTO- National Organ and Tissue Transplant Organisation, ROTTO- Regional Organ and Tissue Transplant Organisation, SOTTO- State Organ and Tissue Transplant Organisation.

¹ Mohan, Lalitha & Selvam, T. (2020). Perspective on organ donation in India: A comprehensive review. *The Journal of Community Health Management.*

² Directorate General of Health Services. *National Organ Transplant Programme*. Retrieved 20 September 2023.

Figure 1 – Key Activities under Organ Donation in India



Key Initiatives Towards Enhancement of Organ Donation in India



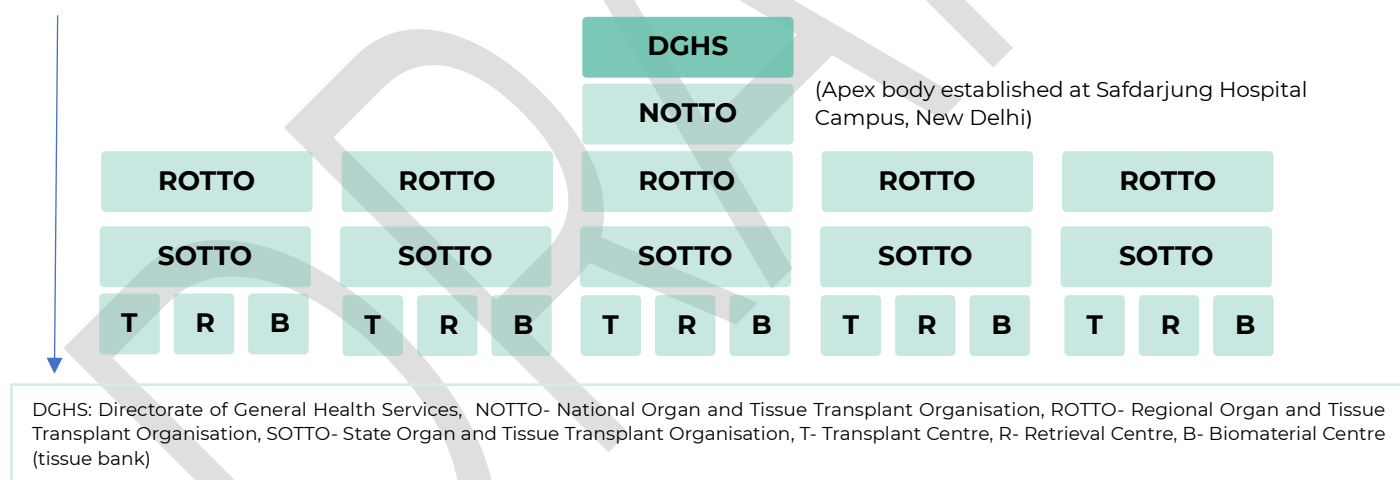
National Organ Transplant Programme (NOTP)

- Government of India introduced NOTP to promote deceased organ donation and address the shortage of organs for transplantation. It aims to increase the availability of organs for transplant and improve the overall organ transplantation landscape in the country.⁴
- Organisations under NOTP aid in coordination and networking for procurement and distribution of organs - NOTTO at the national level, ROTTO at the regional level and SOTTO at the state level.⁴

Objectives of the National Organ Transplant

- Establishing a structured system for organ and tissue procurement and distribution for transplantation.
- Encouraging deceased organ and tissue donation.
- Providing necessary training to the workforce.
- Safeguarding economically disadvantaged individuals from organ trafficking.
- Vigilantly monitoring organ and tissue transplant services and implementing policy adjustments when necessary.

Figure 2 - National Human Organs and Tissues Removal and Storage Network⁵



Note: The above figure is just a sample indicative of the hierarchy of NOTTO, ROTTO and SOTTO and does not equal to the actual number of ROTTOs and SOTTOs available in India



Green Corridors for fast transportation of donated

In 2014, the Government of India introduced the concept of “Green Corridors,” establishing dedicated and cleared road routes for the ambulances transporting donated organs for transplant. These corridors ensure timely and efficient transport, ultimately saving lives by expediting organ delivery to recipients in need.⁶

³ First Organ and Tissue Transplants. Donate Life. Retrieved 23 October 2023.

⁴ Department of Health & Family Welfare.(2023).Annual Report.

⁵ Directorate General of Health Services. (ND). National Human Organs and Tissues Removal and Storage Network.

⁶ Mann Ki Baat. (2023). Organ Donation- A Gift of Life.



One Nation, One Policy for Organ Donation

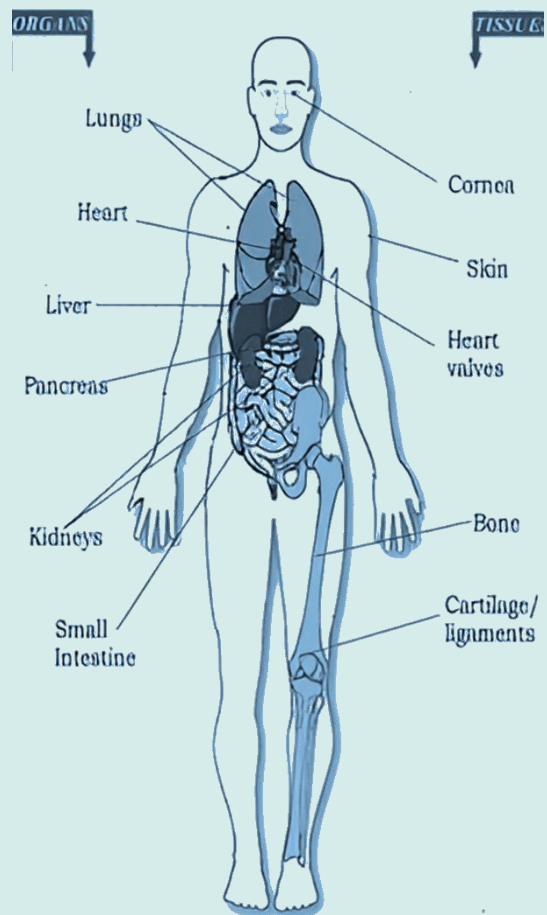
In 2023, the government introduced the “One Nation, One Policy” initiative for organ donation and transplantation in India which enables all citizens to register on a single waiting list, regardless of their domicile. The removal of age limit and the focus on promoting cadaver donation which aims to enhance equitable access to organs, addressing the current shortage in organ transplants.⁶

Organs and Tissues That Can Be Donated

Vital Organs like heart, liver, kidneys, intestines, lungs, and pancreas can be donated only in case of 'brain death'. However other tissues like corneas, heart valves, skin, bones etc can be donated only in case of natural death.

- Types of donors are⁹:
 - **Near related donors:** Which includes immediate family (mother, father, son, daughter, brother, sister and spouse).
 - **Other than near related donors:** Who can donate only out of affection and attachment or only for any special reason with the approval of authorisation committee.
 - **Swapping:** By swapping of near relative donors between pairs of unmatched donor and recipient.
 - **Deceased donors:** Organ Donation is practically possible in the situation of **Brain stem death** (for example, a victim of road traffic accident or cerebrovascular accidents etc.) where the brain stem is dead and person cannot breathe on his own but can be maintained through ventilator, oxygen, fluids etc. to keep the heart and other organs working and functional. After cardiac death only tissues are donated.²
- **Most commonly transplanted organs** include kidney, liver, heart, lung, pancreas and intestines, tissues such as corneas, heart valves, skin and bone.²

Figure 3 – List of Organs and tissues that can be donated⁶



Key Trends in Organ Donation

Organ donation has witnessed several key trends that has shaped its landscape over time and impacted the lives of countless individuals in need. In recent years, these trends have not only evolved but have also become increasingly crucial in addressing the growing demand for organs in a world where shortages persist. Below are the essential trends that influence organ donation, highlighting their impact on the process and the ongoing efforts to enhance its efficiency and accessibility.



1. Increase in Organ Donation

- There has been a gradual increase in organ donation in India. The awareness and acceptance of organ donation have been growing leading to **more than 16,000 donations in 2022**.⁸
- Deceased organ donation rates have improved from 0.27 to 0.52 per million population (2013–2019) with an overall increase of transplant rates from 4990 to 12 666 during the same time period.

Figure 4 – Number of Organ Transplants*

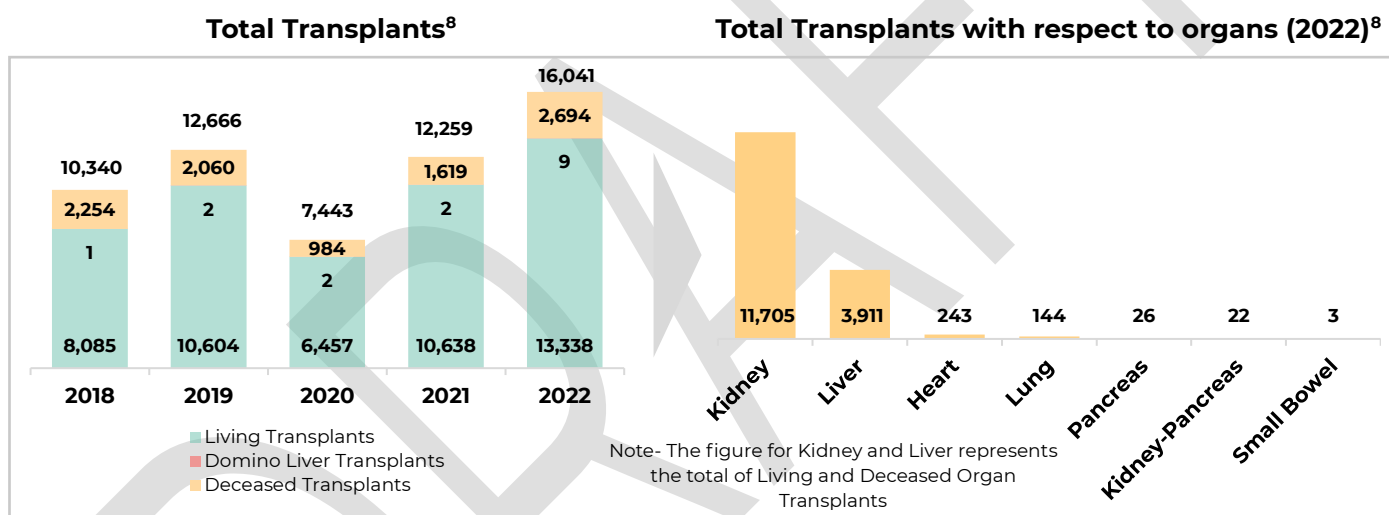
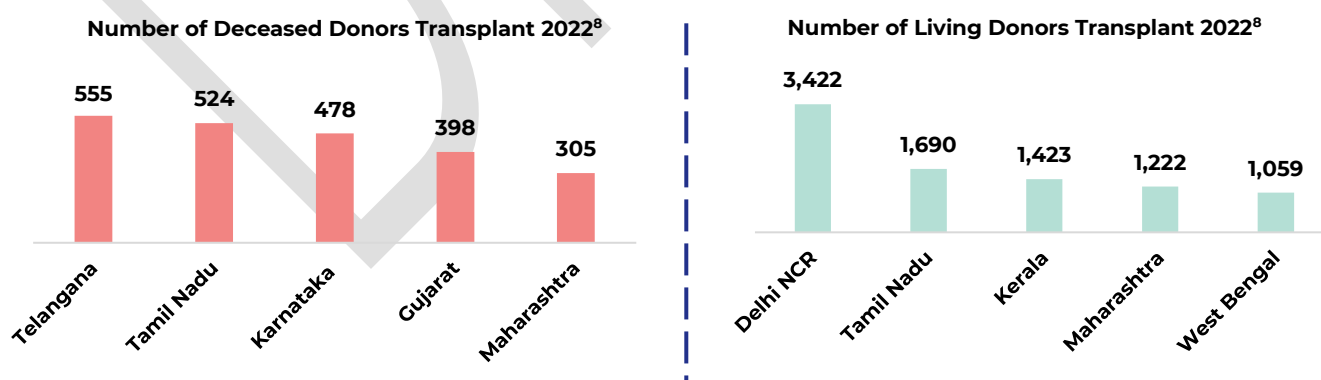


Figure 5 - Top 5 States: Number of Donor Transplants in 2022*



*Note: All the information pertaining to number of organ transplants done in the year 2022 with respect to organ type has been collated into one chart from DGHS website

7 Parada-Contzen et al. (2019). An analysis of economic incentives to encourage organ donation: evidence from Chile. Lat Am Econ Rev.

8 National Organ & Tissue Transplantation Organisation.(2022) *Data for Organ Donation and Transplantation*

9 National Organ & Tissue Transplantation Organisation. (2021). *Guidelines for implementation of National Organ Transplant Programme.*

Key Highlights

- Total number of transplants in 2022 were 16,041 growing from 4,990 in 2013, showing more than **3 times growth**.⁸
- Annual **increase of 30%** was seen in Organ Transplantation in **2022 in comparison to 2021**.⁸
- The most frequent organ transplants were performed for kidney and liver.
- In Deceased Donor Transplants, Telangana is outperforming other states in terms of number of transplants.⁸
- In Living Donor Transplants, Delhi NCR is outperforming other states in terms of number of transplants and does more than Tamil Nadu and Maharashtra cumulatively.⁸
- According to Global Observatory on Donation and Transplantation (GODT), **India ranked 3rd in the world in 2021 after USA and China** in terms of annual number of transplants.⁹



2. Organ Transplant Infrastructure

- Establishment of organisations such as the **NOTTO, ROTTO and SOTTO** has helped in streamlining organ transplantation activities, increasing number of transplant centres, and maintaining registries for organ allocation.²
- Improved coordination facilitated by NOTTO, ROTTO and SOTTO has led to a higher utilisation of organs per deceased donor. In 2016, **2,265** organs were utilised from 930 deceased donors, whereas in 2022, **2,765** organs were utilised from 904 deceased donors.^{8,11}



3. Public Awareness Campaigns

- Promotion of organ donation encompasses a comprehensive approach, including advocacy by leaders and celebrities, inter-ministerial coordination, simplifying pledge processes, including organ donation concept in medical course curriculum, media outreach, NGO involvement, collaboration with medical institutions, CSR initiatives, and stakeholder sensitisation.⁹
- Efforts are being made with regional campaigns and national levels to drive awareness about organ donation.²



4. Organ Transplantation Centres

- Specialised transplant centres have emerged in various parts of India, providing comprehensive organ transplantation services. There are a **total of 685 registered hospitals** performing organ and tissue transplantation as of 2022.²
- Tamil Nadu has its own government society, Transplant Authority of Tamil Nadu (TRANSTAN). It functions as a ROTTO for the southern region and SOTTO as per the government orders and conducts various activities including streamlining all procedures related to organ transplantation.¹⁰

Recently in March 2023, Prime Minister Narendra Modi in one of his "Mann Ki Baat" programmes urged the citizens to actively participate in the noble cause of Organ donation⁶

¹⁰ Transplant Authority of Tamil Nadu. About TRANSTAN. Retrieved 20 September 2023.

¹¹ MOHFW. (2023). Brief Activities of MOHFW.

Opportunities and Challenges

Opportunities



Spreading Awareness and Educating Public

India's organ donation rate (ODR) is quite low (0.34 per million populations) in comparison to other countries. High level of awareness is usually associated with a higher rate of organ donation.



Developing Transplant Infrastructure

Establishing more transplant centres, organ retrieval facilities, and well-equipped intensive care units (ICUs) in hospitals to support deceased organ donation and transplantation.



Streamlining Efforts

Improved coordination among government bodies, healthcare institutions, NGOs, professional organisations, and community leaders can lead to success and better awareness in organ donation initiatives.⁹



Emulating successful practices from states

Good practices examples can be taken from states like Tamil Nadu and Maharashtra.

For example, in Tamil Nadu, streamlining the organ distribution process, clarifying medical legal procedures and making the declaration of brain death is mandatory in government medical college hospitals.

Establishment of a customised UNOS (United Network for Organ Sharing) model, as implemented in Maharashtra through ZTCCs (Zonal Transplant Coordination Centres), with a focus on achieving financial sustainability.^{10,12}

¹² Directorate General of Health Services. (2022). *Organ Donation and Transplantation*

¹³ Patel et al. (2018). *Deceased Organ Donation in India with Indian Journal of Transplantation*.

Challenges



High Patient Burden

There is a wide gap between patients who need transplants and the organs that are available in India. For example, an estimated around 1.8 lakh persons suffer from renal failure every year, however the number of renal transplants done is around 6000 only.²



Limited infrastructure and Organisational Support

Limited infrastructure, particularly within government sector institutions, in the context of organ donation and transplantation, is a notable concern.⁹

There is a need for more transplant centres, along with adequately educated transplant coordinators and staff who are skilled in the essential procedures for executing organ donation initiatives.¹³



Limited Brain Stem Death Identification

Brain Stem death is recognised as a legal death in India under the Transplantation of Human Organs Act, however more awareness is required to be built on **Brain Stem Death Identification and Certification.**²



Organ trafficking

Organ trafficking and commercialisation remain serious concerns in organ donation. Illicit activities and unethical practices undermine the integrity of organ donation programs and pose significant challenges to the ethical procurement and allocation of organs.²



Limited awareness about organ donation procedures

A significant barrier to organ donation in India arises from the unawareness about registration process of organ donation pledges, which highlights the need to improve the campaigns for registration.

Cross Cutting Themes



Education and Awareness

Significance: Education and awareness are crucial in reducing stigma, and encouraging organ donation. Informed citizens can make voluntary, informed decisions about donation, promoting a culture of giving.

Current Efforts: Government has initiated several awareness campaigns like: **“Jeete ji Raktdaan, Mrityu ke baad Angdaan”, “Har Ghar Hai Donor”, Online donation pledge registry in NOTTO website etc.** are being conducted to educate and promote organ donation amongst public.

Way Forward: Emphasize organ donation education in school curricula, expand public awareness campaigns, and engage the media and celebrities to reach a wider audience.



Health Systems Strengthening

Significance: A robust healthcare infrastructure is critical to ensure the safe, efficient, and equitable transplantation process. Strengthened health systems improve access and post-transplant care.

Current Efforts: The government has taken steps to establish **NOTTTO, SOTTO, and ROTTO** to ensure a fair and organized system for organ donation supply and management.

Way Forward: Invest in transplant-specific digital infrastructure.

Ministries and Departments

Inter-ministerial and departmental collaboration is an important aspect of governance in India. Various ministries, departments, and organisations work together to foster a culture of organ donation and ensure efficient allocation and distribution to those in need. These collaborative efforts are essential to save and enhance lives through the invaluable gift of organ donation.

Table 1 – Ministries and Departments working towards Organ Donation in India

 Ministry	 Department and Bodies	 Functions and Roles
 Ministry of Health and Family Welfare	 Department of Health and Family Welfare  Directorate General of Health Services	- Ensuring the availability and accessibility of quality healthcare services to all citizens through government hospitals, health centres, and other healthcare facilities. - It mainly functions in implementing health policies, family planning programmes, disease surveillance, Health Promotion and Education etc. - Oversees operations and management of government-run hospitals at the central level. Additionally, it handles people's health issues through its affiliated offices and institutes located across the nation. - NOTTO, ROTTO and SOTTO are National, Regional and State level organisations set under this department for coordination of Organ and Tissue donation and transplantation activities.
 Ministry of Law and Justice	 Department of Justice	The Department plays a role in organ donation by providing legal guidance and ensuring that organ transplantation activities are being conducted adhering to the relevant laws and regulations.

Government Interventions

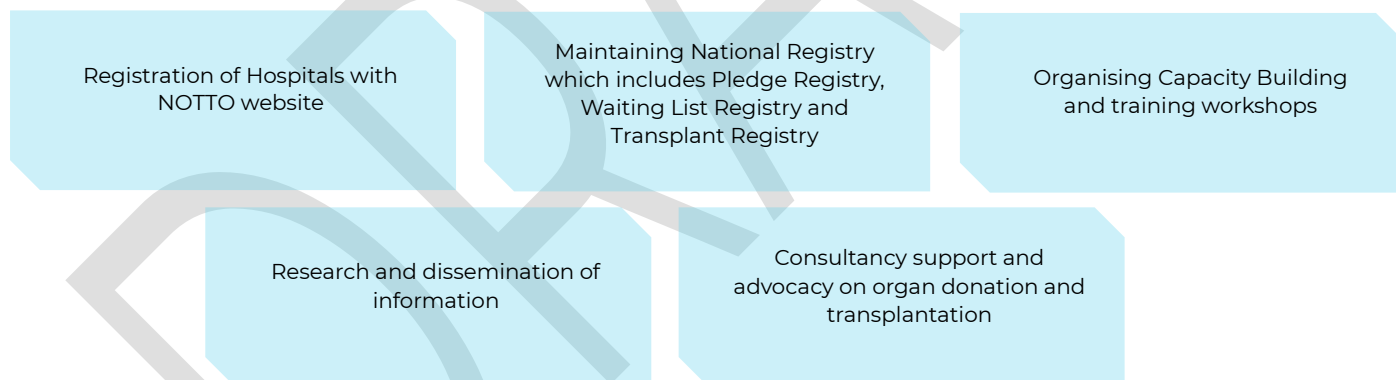
A. National Organ and Tissue Transplant Organisation (NOTTO)⁴

NOTTO serves as the central authority responsible for the coordination, networking, and oversight of organ and tissue transplantation activities at the national level in India. It maintains the National Organ and Tissue Transplant Registry (NOTTR) to maintain a record of all the transplant activities conducted in the country.

Main Functions



Main Activities



Key Achievements⁴



implemented both online and offline organ donation pledging mechanisms.

16,58,257 Pledges received as of Sept,2022



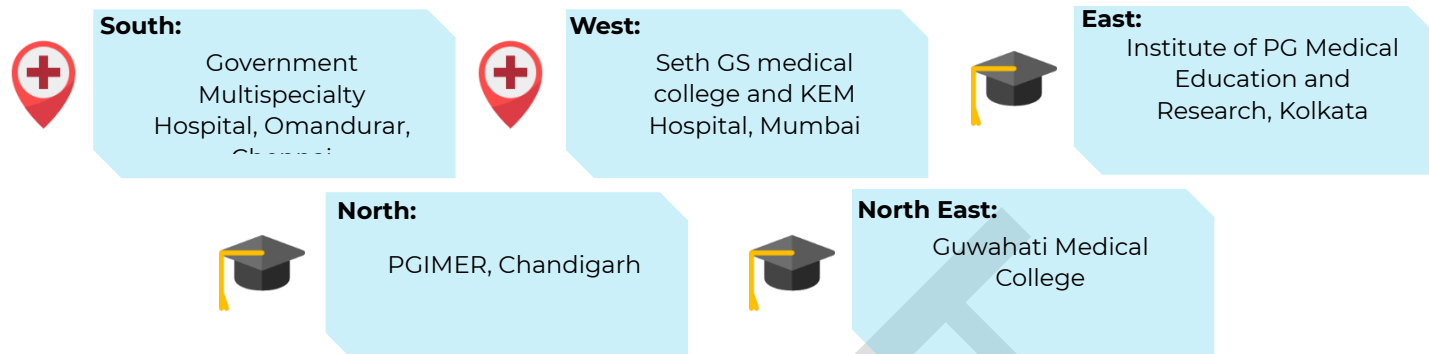
685 Hospitals Registered with NOTTO website



establishment of a 24/7 call centre with a toll-free helpline number (1800114770), for providing better information in facilitating network of hospitals

B. Regional Organ and Tissue Transplant Organisation (ROTTTO)⁴

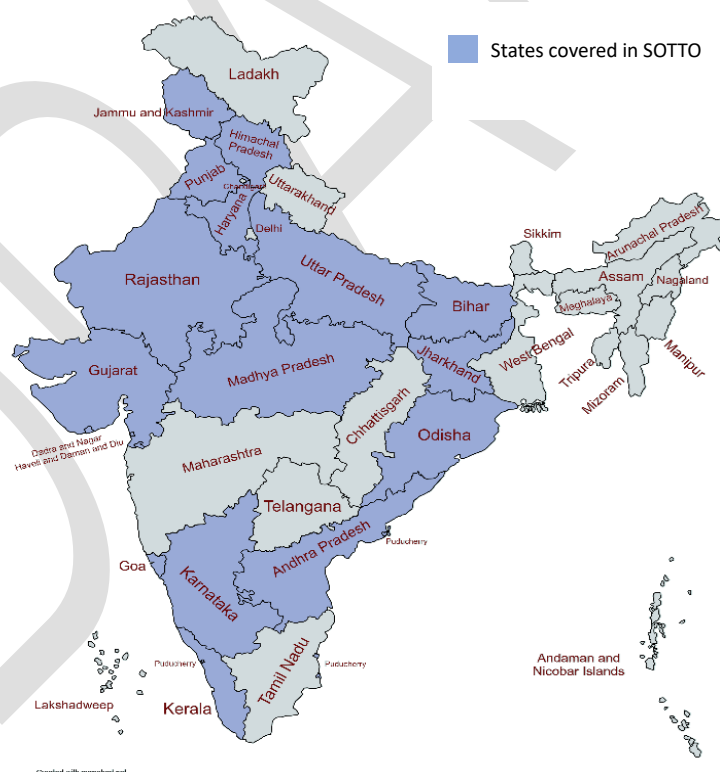
ROTTTO operates at the regional level, covering a specific geographic region within a state. There are currently 5 ROTTOs who serve their respective regions.



C. State Organ and Tissue Transplant Organisation (SOTTO)⁴

SOTTOs operate at the state level and serve as the nodal agencies for organ and tissue transplantation activities within their respective states. There are currently **21 SOTTOs**¹¹ sanctioned by Government of India (GoI).

Figure 6 – SOTTO coverage in India



The main focus of NOTTO, ROTTO and SOTTO is on enhancing infrastructure and increasing the availability of trained healthcare professionals for organ transplantation especially in government institutions and to facilitate all states and union territories to upload their organ transplant data on the web portal of the NOTTO which is to be upgraded to improve data management and accessibility.⁴

D. One Nation, One Policy for Organ Donation⁶

In 2023, Government of India adopted "One Nation, One Policy" for Organ Donation and Transplantation. It eliminates the state domicile requirement and the upper age limit of 65, ensuring equitable access to organ transplantation for all individuals across the country, can also be used for digital allocation of organs.

"In this era of modern medical science, organ donation has become a very important means of imparting life to someone. It is said that when a person donates one's body after death, it creates a possibility for 8-9 people to get a new life."

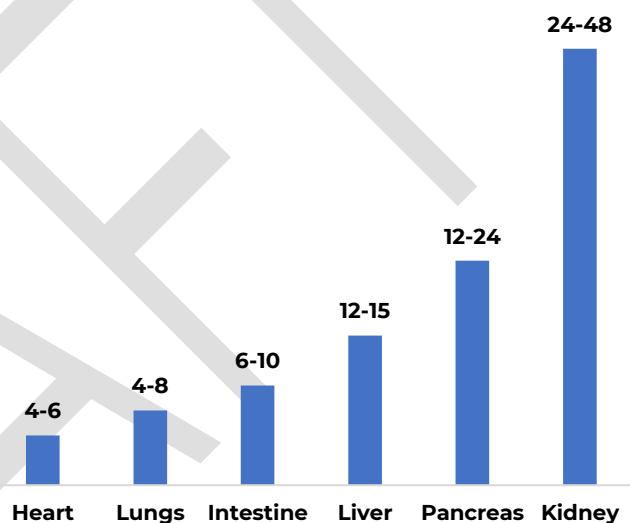
- Prime Minister Narendra Modi (in his 'Mann ki Baat' address)⁶

E. Green Corridors¹⁴

In 2014, Government introduced the concept of "Green Corridors", which is a dedicated and unrestricted route specifically created for ambulances transporting harvested organs for transplantation to reach the designated hospital. It ensures rapid and uninterrupted transport, allowing the organs to reach their destination within the shortest possible time.

The establishment of green corridors is crucial because organs have limited preservation time, and minimising the duration between harvest and transplant increases the chances of a successful surgery. Therefore, early transportation plays a significant role in enhancing the success rates of organ transplantation.

Figure 7 – Transplantation time (in hours) for various organs



F. Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY)^{11,15,16}

The Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (AB-PMJAY) is a flagship health insurance scheme launched by the Government of India. It is aimed at providing financial protection and access to quality healthcare services to economically vulnerable families across the country.

Organ and Tissue transplants is one of the benefits in the National Health Benefit Package (HBP) under Ayushman Bharat Pradhan Mantri Jan Arogya Yojana (PM-JAY). It consists of services of renal transplants and corneal transplants. There is one package and a total of 6 procedures for Organ and Tissue Transplant and the total package price is yet to be announced according to Health Benefit Package 2.2.

Jammu and Kashmir is one of the first few states that have performed an organ transplant covered under Ayushman Bharat PMJAY in 2022.

¹⁴ Koushal V et al. (2018). *Impact of green corridors in organ donation with Indian J Transplant.*



Building Awareness⁹

Implementation of awareness programmes by building large scale campaigns educating the people about the importance of organ donation. People can be educated about brain stem death, process of organ donation, and simplifying the pledging process for Organ Donation. It can be done with the help of leaders and celebrities, social media, television, promotion of organ donation as a corporate social responsibility and other sources which can reach a wider audience.



Grief Counsellors¹⁷

To encourage and motivate families for organ donation, counselling on various positives of organ donation such as saving of lives, moral obligation, being a role model for others, etc can help. Also, by honouring the generosity of donor families and recognising and appreciating their efforts can help in fostering a positive organ donation culture and encourage more individuals to consider organ donation.



Increasing Transplant Coordinators¹²

Transplant Coordinators play a pivotal role throughout the organ donation process. Increasing the number of coordinators will help in creating awareness and educating the relatives the donation process. Provision and training of transplant coordinators in the emergency room and ICU help in initiating the grief counselling for the family members for organ donation.



Updating communication and awareness strategy⁹

Spreading information about the government guidelines for donation after brain stem death and use of new campaigns to encourage organ donation among the changing demographic of the country.



Capacity building for medical Institutions⁹

Expanding the availability of transplantation services to more public facilities through sensitization and training of healthcare professionals.



Digital Registry on organ Transplantation

NOTTO along with Ayushman Bharat, has launched an online registry which gives live update on Total Pledge Registrations for organ donation. Access to such real time information will bring more transparency.

¹⁵ National Health Authority. (2021). *Guidelines for National Health Benefit Package 2.2*.

¹⁶ Sher-I-Kashmir Institute of Medical Sciences Soura, Srinagar. *AB-PMJAY*. Retrieved 20 September 2023.

¹⁷ Saxena D et al. (2022). Challenges and Motivators to Organ Donation: A Qualitative Exploratory Study in Gujarat, India. *Int J Gen Med*.

Best Practices

1. Transplantation Authority of Tamil Nadu (TRANSTAN)¹⁸



Background

Cadaver Transplant Programme of Tamil Nadu was initiated in 2014 by the Government of Tamil Nadu that focuses on promoting deceased organ donation and facilitating organ transplantation procedure. TRANSTAN coordinates and supervises the organ donation procedure for living, cadaver and tissue transplants.



Steps taken by the government

- The government provided financial assistance to people in need through the Chief Minister's Comprehensive Health Scheme up to 22 lakhs.
- The state streamlined the organ allocation process by which the potential organ donors can be matched with the suitable recipient .
- Encouraged new hospitals to join the programme by providing assistance.
- Collaborated with hospitals in the state to facilitate organ retrieval, transplantation surgeries and post-operative care.
- The government made the certification of brain stem death mandatory.
- Conducted various awareness campaigns for the promotion of Organ Donation in the state some of which are:
 - Organ Donation Week- Awareness rally
 - Awareness Stall of TRANSTAN
 - Transplant Coordinators Training Programme
- Launched the website- TRANSTAN for transplant registry.

Figure 8 – Awareness stall of TRANSTAN



Impact generated

- Tamil Nadu became the first state to have performed renal transplants in 13 government medical colleges and hospitals and 23 government Non-Transplant Organ Retrieval Centres (NTORC) hospitals.
- Clarified Medical Legal Procedures related to Organ Donation from Brain Dead Donors.
- Complex procedures like transplantation of Pancreas and Kidney, Liver and Kidney, Heart and Lung were undertaken.
- Expansion of the forensic team for the transplantation procedure.
- Many states and even the Government of India have replicated the model of Tamil Nadu in their own states for better management of the procedure.

¹⁸ National Health Mission. (ND)TRANSTAN & Organ Donation activities in Tamil Nadu.

- Total transplantations conducted as of 30th October 10,003, including 777 Heart transplants, 785 Lung transplants, 1,538 Liver transplants, 2,986 Kidney transplants, 2,450 Cornea, and others.¹⁶

2. Zonal Transplant Co-ordination Centre (ZTCC), Mumbai (Maharashtra)²⁰

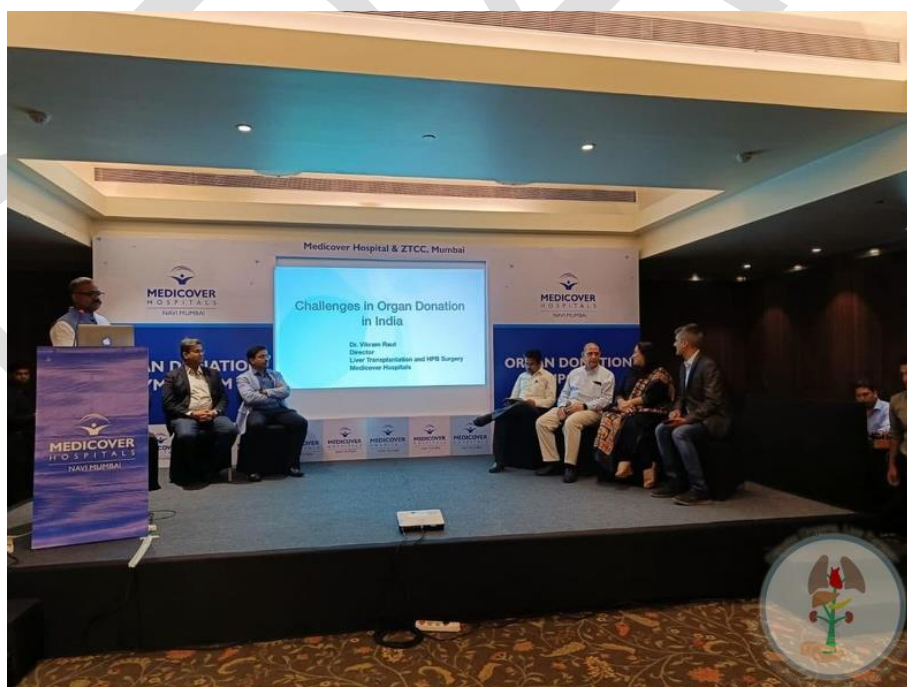
Background

The Zonal Transplant Co-ordination Centre was set up with the aim of coordinating, monitoring and supervising the cadaver organ transplant programme and tracking the organ distribution in the state. It also aims to promote the recognition and awareness of organ donation.

Steps taken by the government

- Maintaining common waiting list for the recipient in need of organ for each organ at the city level.
- Matching the suitable donor and receiver of organ according to the state guidelines.
- Formation of experts committee council for technical guidance.
- Controlling the organ distribution in the city.
- Generating awareness for organ donation by organising awareness campaigns/programmes for both professionals and the general public educating about deceased organ transplantation, some activities are:
 - Community Meetings.
 - Awareness Workshops.
 - Conducting marathons, rallies.
 - Facilitation of donor's family members.
 - Pamphlet distribution.
 - Conducting training programmes for sensitisation of professionals and para-professionals.

Figure 10 – Organ Donation Symposium at ZTCC



²⁰ Zonal Transplant Co-ordination Centre, Mumbai. Retrieved 20 September 2023.

Impact generated

- Multiple awareness programmes were successful and could make an impact about the awareness of organ donation in the city.
- It has been made mandatory to declare Brain Death to the ZTCC.
- Total transplants done in Mumbai from 1997 till 30th October, 2023 are Kidney – 1,014; Liver – 472 and Heart – 193, and Lung- 41.¹⁸

Recommended Readings

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